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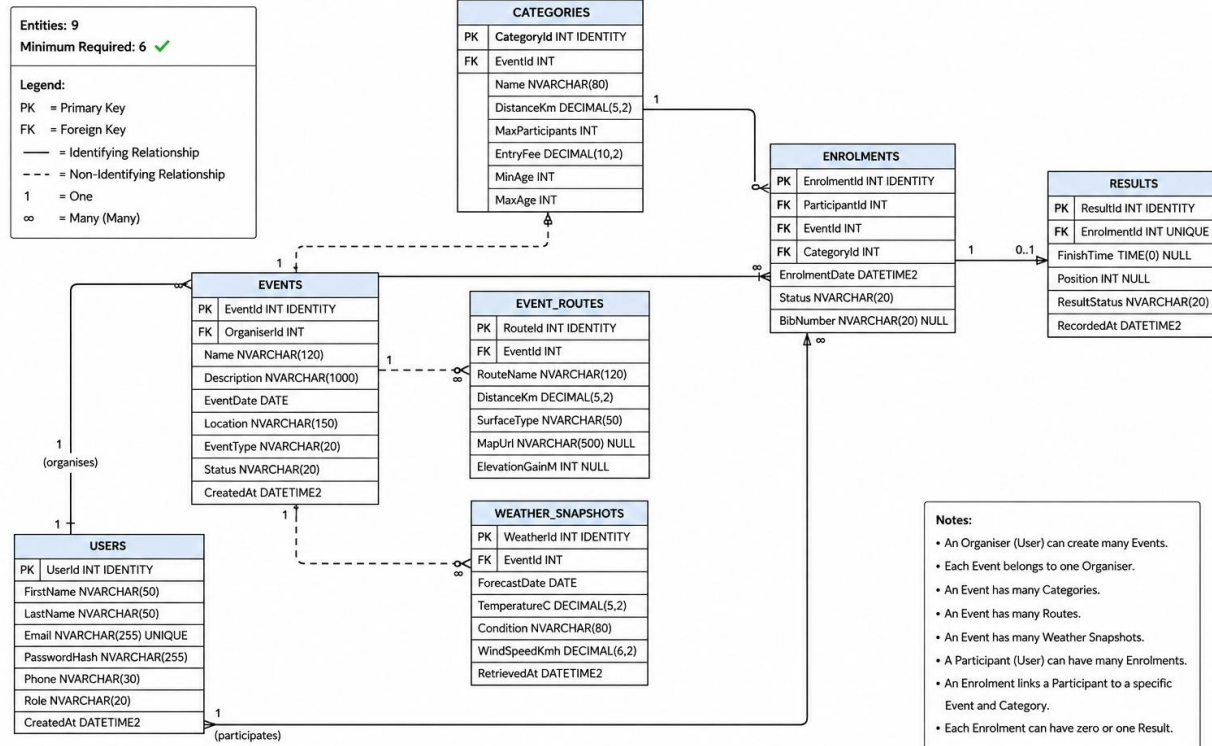
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PROG6121 POE Part 1



1. Entity Relationship Diagram (ERD)

RaceDay System – Entity Relationship Diagram (ERD)



1. What is an ERD?

ERD stands for **Entity Relationship Diagram**.

It shows:

- **Entities** → the database tables, e.g. USERS, EVENTS, CATEGORIES
- **Attributes** → the columns inside each table
- **PK (Primary Key)** → uniquely identifies each record
- **FK (Foreign Key)** → connects one table to another
- **Cardinality** → explains how many records in one table can be related to records in another

Your system has these main entities:

1. USERS
2. EVENTS

- 3 . CATEGORIES
- 4 . EVENT_ROUTES
- 5 . WEATHER_SNAPSHOTS
- 6 . ENROLMENTS
- 7. RESULTS

The ERD shows how the RaceDay system stores and connects its data. Users can be either Organisers or Participants. Organisers create events, and each event can have multiple categories and weather snapshots, as well as an optional route. Participants enter events through enrolments, where they select a category (IBM,2024). Each enrolment can then have zero or one result recorded after the race. Primary and foreign keys connect the tables and maintain the relationships between them.

Cardinalities

The main cardinalities in the RaceDay ERD are:

Relationship	Cardinality	Explanation
Users → Events	1 : Many	One organiser can create many events.
Events → Categories	1 : Many	One event can contain many categories.
Users → Enrolments	1 : Many	One participant can have many enrolments.
Events → Enrolments	1 : Many	One event can have many participants/enrolments.
Categories → Enrolments	1 : Many	One category can have many enrolments.
Events → Event Routes	1 : 0..1	An event can have zero or one route.
Events → Weather Snapshots	1 : Many	One event can have many weather snapshots.
Enrolments → Results	1 : 0..1	An enrolment can have zero or one result.

2. API Endpoint Plan

This endpoint plan defines the REST API for the RaceDay system. It is based on the database ERD and the functional requirements (IBM,2025).The API uses JSON request/response bodies and enforces the two application roles: Organiser and Participant.

Roles

- **Public** – no authentication required.
- **Any logged-in user** – authentication required.
- **Participant** – authenticated user whose role is Participant.
- **Organiser** – authenticated user whose role is Organiser.

The RaceDay system uses a **REST API** to allow the client application to communicate with the server and access resources such as users, events, categories, enrolments, and results. REST APIs use HTTP methods such as GET, POST, PUT and DELETE to retrieve, create, update and remove resources (IBM, 2025).

Authentication

HTTP METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
POST	/api/auth/register	Creates a new user account so a participant can access the RaceDay system.	Public	{ email, password, fullName, role }	201 Created – account created. 400 Bad Request – invalid/missing data. 409 Conflict – email already registered
POST	/api/auth/login	Authenticates an existing user and returns authentication information including their role.	Public	{ email, password }	200 OK – login successful with token/session and user details. 400 Bad Request – invalid request. 401 Unauthorized – incorrect credentials.
GET	/api/users/profile	Get current user's profile	Any	None	200 OK – user details
PUT	/api/users/profile	Update current user's profile	Any	{ fullName, email? }	200 OK – updated user 400 – validation errors
GET	/api/events		Public		

		List all events (with optional filters: status, date range)		None (query params)	200 OK – array of event summaries
POST	/api/events	Create a new event	Organiser	{ name, description, date, venueId, maxParticipants, status }	201 Created – full event details 400 – validation
GET	/api/events/{id}	Get detailed information for a specific event	Public	None	200 OK – event with categories 404 Not Found
PUT	/api/events/{id}	Update an existing event (only organiser of that event)	Organiser	{ name, description, date, venueId, maxParticipants, status }	200 OK – updated event 403 Forbidden – not organiser
DELETE	/api/events/{id}	Delete an event (only organiser)	Organiser	None	204 No Content 404 – not found
GET	/api/events/{id}/categories	Get all categories for an event	Public	None	200 OK – array of categories
POST	/api/events/{id}/categories	Add a new category to an event	Organiser	{ name, description, entryFee }	201 Created – category details
PUT	/api/categories/{id}	Update a category (organiser of parent event)	Organiser	{ name, description, entryFee }	200 OK – updated category
DELETE	/api/categories/{id}	Delete a category (organiser of parent event)	Organiser	None	204 No Content
GET	/api/events/{id}/enrollments	List all enrollments for an event (with results)	Organiser	None	200 OK – array of enrollment details
POST	/api/events/{id}/enroll	Enroll the current participant in an event	Participant	{ categoryId }	201 Created – enrollment details 400 – event full, duplicate, etc.
GET	/api/enrollments/me	Get all enrollments	Participant	None	200 OK – array of

		of the current participant			enrollments with event/category
PUT	/api/enrollments/{id}	Update enrollment status (e.g., cancel)	Participant (own)	{ status }	200 OK – updated enrollment
POST	/api/enrollments/{id}/result	Record a result for an enrollment	Organiser (of parent event)	{ timeInSeconds , overallPosition , categoryPosition }	201 Created – result details 409 Conflict – result already exists
GET	/api/enrollments/{id}/result	Get result for a specific enrollment	Any (or public?)	None	200 OK – result details 404 – not found
GET	/api/events/{id}/results	Get all results for an event (leaderboard)	Public	None	200 OK – array of results with participant names
GET	/api/results/me	Get current participant's personal results history	Participant	None	200 OK – array of results with event info

Reference list

IBM (2024) *What is an Entity Relationship Diagram?* Available at: [IBM – Entity Relationship Diagram](#) (Accessed: 1 September 2026).

IBM, 2025. *What is a REST API (RESTful API)?* [online]. Available at: [IBM REST API explanation](#) [Accessed 1 September 2026]

IBM, 2024. *What is an API (Application Programming Interface)?* [online]. Available at: [IBM API overview](#) [Accessed 1 September 2026].